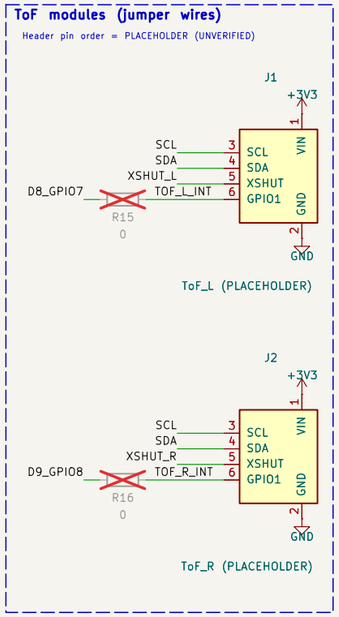
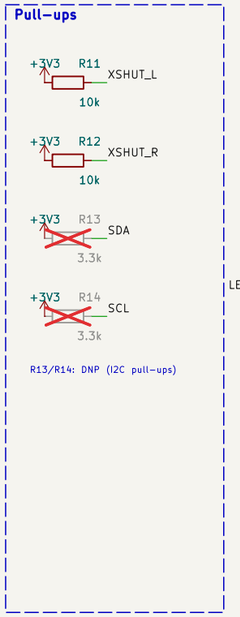
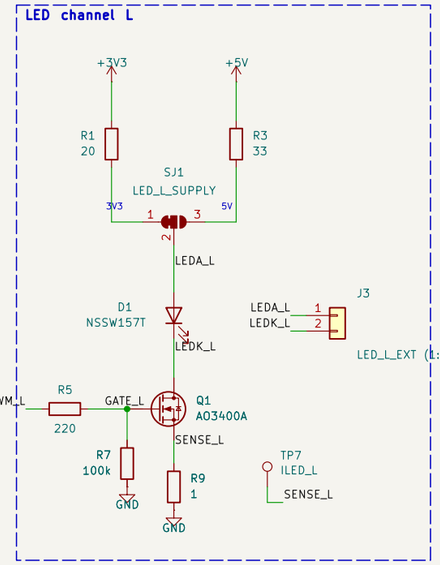
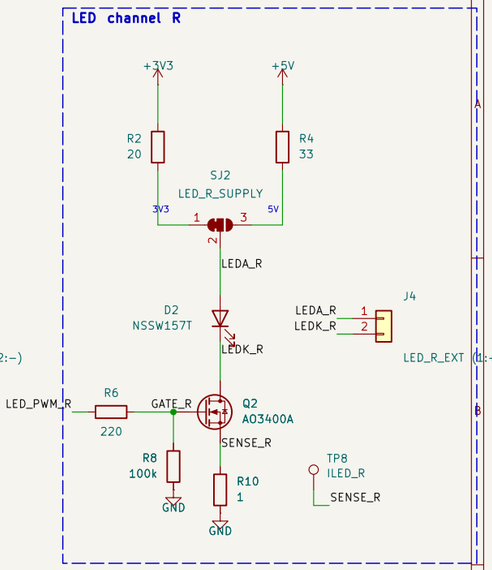








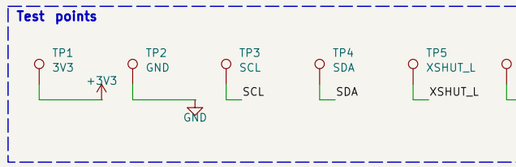


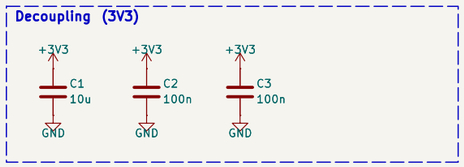
F. GPIO assignment (approved 2026-10-01)

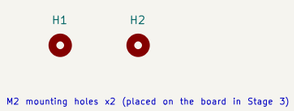
D0=GPIO1 XSHUT_L | D1=GPIO2 XSHUT_R | D2=GPIO3 LED_L PWM (strapping pin, gate pulled low) | D3=GPIO4 LED_R PWM

D4=GPIO5 SDA | D5=GPIO6 SCL | D6/D7=GPIO43/44 reserved UART | D8/D9=GPIO7/8 = ToF GPIO1 via DNP links R15/R16 (shared with Sense microSD) | 5V: SJ option only

Firmware: drive both XSHUT low first, release one at a time and change the VL53L1X address (default 0x29 7-bit = 0x52 8-bit).







NOTES (full calculation: docs/spec.md, tools/led_budget.py)

A. POWER

Everything is powered from the XIAO 3V3 pin. XIAO 5V pin is wired ONLY to R3/R4 -> solder jumper pad 3 (open by default).

LED supply per channel: SJ1/SJ2 default = pads 1-2 bridged = 3V3 via R1/R2 (20 ohm).

For 5V: cut the 1-2 bridge, bridge 2-3 (R3/R4 33 ohm 2010 are already fitted). 5V = USB VBUS (no voltage on battery power).

B. LED @3V3 (R1/R2 20 ohm 0603)

Vf(I) = 2.677 V + 5.29 ohm*I (Nichia STS-DA1-1913: 3.1 V typ and 2.8-3.4 V @80 mA; 2.73 V @10 mA read from the graph = reference only).

Vcc 3.2-3.4 V (assumed), Rds(on) <= 75 mohm. Per LED min/nom/max = 8.5 / 23.6 / 38.8 mA

= 6 / 16 / 26 % of IF max 150 mA = 11 / 30 / 49 % of the 80 mA rated flux.

R1/R2 dissipation <= 30 mW (30 % of 0.1 W). Both LEDs at 100 % duty, max: 77.6 mA.

C. LED @5V (R3/R4 33 ohm 2010 0.5 W)

Vcc 4.75-5.25 V: 45.0 / 59.0 / 73.0 mA per LED (49 % of IF max). R3/R4 dissipation <= 176 mW (35 % of 0.5 W).

5V total with the XIAO webcam capture peak 347 mA (Seeed wiki) = 493 mA (< 500 mA USB 2.0). XIAO 5V pin current capability: UNVERIFIED.

With the wrong resistor (20 ohm on 5V) a LED would get 109 mA and R1 238 mW -> always keep SJ and R1/R3 as drawn.

G. UNVERIFIED / WARNINGS

ToF header pin order is a PLACEHOLDER (module not in hand). Gerbers are NOT released until the real pinout is checked. ToF modules are wired with jumper wires.

J3/J4 external LED: pin 1 = + (after the series resistor), pin 2 = -. Fit EITHER D1/D2 or an external LED (both = they share the current).

Do NOT short J3/J4 + to -: R1 would dissipate 0.49 W (0603 = 0.1 W). R3 0.71 W (2010 = 0.5 W).

D. 3V3 TOTAL (Seeed: 'You can draw 700mA', condition not stated)

External: LEDs 77.6 + ToF peak 2 x 40 + XSHUT pull-ups 0.7 = 158 mA (81 mA if the LEDs are on 5V).

XIAO Sense (Seeed wiki): webcam average 140 mA, capture peak 366 mA -> total 524 mA (75 % of 700 mA).

Seeed camera table MAX ~0.65 A (conditions not stated = UNVERIFIED) -> 808 mA (115 %). ESP32-S3 Wi-Fi TX peak 340 mA (chip only, Datasheet v2.2 Table 5-7).

=> Measure the real 5V input current before running Wi-Fi TX + capture + LEDs 100 % + ToF peak together.

E. GPIO and MOSFET gate drive

GPIO carries no LED current (gate only). Gate charge peak 3.3 V / 220 ohm = 15 mA; XSHUT low sinks 0.33 mA. ESP32-S3: IOH 40 mA / IOL 28 mA (Datasheet v2.2 Table 5-4).

3.3 V gate drive is enough: ESP32-S3 VOH >= 0.8*VDD = 2.64 V. AO3400A (AOS Rev 3.1): VGS(th) 0.65-1.45 V, Rds(on) < 48 mohm @2.5 V.

SI2302CDS (Vishay S12-2336 Rev D): VGS(th) 0.40-0.85 V, Rds(on) < 75 mohm @2.5 V, VGS max +-8 V. Both SOT-23, pin 1=G 2=S 3=D.

H. OPERATING CAUTIONS (user, 2026-10-01)

5V is a TEST OPTION only. The default is 3V3 (SJ1/SJ2 pads 1-2 bridged).

The 5V total of 493 mA is right at the USB 2.0 limit of 500 mA (margin 7 mA).

3V3: do NOT run Wi-Fi TX, camera capture, LEDs at 100 % and ToF peaks at the same time, until the real 5V input current of the XIAO has been measured.

ToF pin order: PLACEHOLDER (UNVERIFIED)
Power: XIAO 3V3; LED 3V3 or 5V by solder jumper
Prototype for T6 (ToF cliff) / T4 (LED) tests
Serpens / Floor Watch

Sheet: /		
File: head-sensor-board.kicad_sch		
Title: head-sensor-board		
Size: A3	Date: 2026-10-01	**Rev: A**
KiCad E.D.A. 10.0.1		Id: 1/1